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Detection system and detection method for detecting tab back of bare battery cell in cell combining apparatus

Abstract

A detection system and method for detecting a tab back of a bare battery cell during a cell combining operation are disclosed. The system includes a top cap carrying apparatus, a cell combining apparatus, a first detection apparatus, a second detection apparatus, and a control apparatus. The cell combining apparatus displaces two bare battery cells from a lying state to an upright cell combining state when a top cap is supported. The top cap carrying apparatus moves from a carrying position to a displacement position, enabling the tabs to extend and expose an anode tab back part and a cathode tab back part. The first and second detection apparatuses capture images of the respective tab back parts. The acquired images are processed for use in subsequent operations. This system enables real-time image acquisition of the tab back parts during the combining process, providing comprehensive quality information for the bare battery cell tabs.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of PCT Application No. PCT/CN2024/113193, filed Aug. 19, 2024, which claims priority to Chinese Patent Application No. 202410146273.0, entitled “DETECTION SYSTEM AND DETECTION METHOD FOR DETECTING TAB BACK OF BARE BATTERY CELL IN CELL COMBINING APPARATUS” and filed on Feb. 2, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a detection system and a detection method for detecting a tab back of a bare battery cell in a cell combining apparatus.

BACKGROUND

(2) A cell combining operation exists in an assembly process of a bare battery cell. To be specific, two bare battery cells are switched from a lying state to an upright cell combining state with tabs facing downward. During cell combining, a body of the bare battery cell needs to be “lifted up” by 90 degrees relative to a welded top cap. Consequently, a surface layer of a tab between the body of the bare battery cell and the top cap is prone to breakage due to a bending force. Especially, a back part of the tab facing outward is prone to cracking. Therefore, a cracking status of the tab back needs to be detected online when the cell combining operation is completed, so as to obtain information about quality of the tab back for use in a subsequent process.

SUMMARY

(3) In view of the foregoing problems existing in the related art, embodiments of the present disclosure are intended to provide a detection system and a detection method, so as to perform online detection on an anode (cathode) tab back part of a bare battery cell during a cell combining operation, thereby detecting whether a defect exists in the tab back part of the bare battery cell, for use in a subsequent process.

(4) In one aspect, the present disclosure provides a detection system for detecting a tab back of a bare battery cell in a cell combining apparatus. The detection system includes a top cap carrying apparatus, a cell combining apparatus, a first detection apparatus, a second detection apparatus, and a control apparatus. The top cap carrying apparatus is configured to be movable at least between a carrying position that supports a top cap of a bare battery cell and a displacement position that exactly abuts against but no longer supports the top cap. The top cap of the bare battery cell is welded together with an anode tab and a cathode tab. The cell combining apparatus is configured to displace bodies of two bare battery cells disposed in parallel from a lying state to an upright cell combining state in a case that the top cap is supported by the top cap carrying apparatus. In the upright cell combining state, the body of the bare battery cell is completely held by a holding

apparatus and a tab back faces outward. The top cap carrying apparatus is configured to move, under a drive of a first drive mechanism, from the carrying position to the displacement position after the bare battery cell is in the upright cell combining state, to allow the anode tab and the cathode tab to extend to expose an anode tab back part and a cathode tab back part within a desired range. The first detection apparatus and the second detection apparatus each include an image acquisition mechanism. The first detection apparatus is configured to perform image acquisition on the anode tab back part laterally. The second detection apparatus is configured to perform image acquisition on the cathode tab back part laterally. The control apparatus is configured to control the cell combining apparatus to flip the body of the bare battery cell from the lying state to the upright cell combining state, control movement of the top cap carrying apparatus, control image acquisition by the first detection apparatus and the second detection apparatus, and receive and process acquired images, to determine whether defects exist in the anode tab back part and the cathode tab back part, for use in a subsequent process.

(5) Therefore, the detection system of the present disclosure implements online detection on the anode tab back part and the cathode tab back part of the bare battery cell by using the mobility of the top cap carrying apparatus, thereby obtaining more comprehensive information of quality of the tab back.

(6) According to one or more embodiments, the top cap carrying apparatus includes a support plate and a first drive mechanism. The support plate is configured to support the top cap of the bare battery cell, and the first drive mechanism is disposed on a platform. The first drive mechanism is connected to the support plate. The first drive mechanism is connected to the support plate, and is capable of driving the support plate to move in a direction from the bare battery cell to the platform, so that the anode tab and the cathode tab extend.

(7) In the foregoing structure, the first drive mechanism can provide a drive force for movement of the support plate, and the support plate can transfer the drive force of the first drive mechanism to the top cap of the bare battery cell. By moving the support plate, the top cap of the bare battery cell can be controlled to move between the top cap carrying position and the displacement position, to adjust extension states of the anode tab and the cathode tab.

(8) According to one or more embodiments, the first drive mechanism further includes a limiting fixture. The limiting fixture is disposed at an end of the support plate facing the bare battery cell. A surface of a side of the limiting fixture facing the bare battery cell is recessed to form an accommodating groove. A shape of the accommodating groove matches that of the top cap of the bare battery cell.

(9) In the foregoing structure, the limiting fixture can guide a movement direction of the top cap of the bare battery cell, thereby reducing a risk that the top cap of the bare battery cell is deformed and displaced in a flipping process, improving detection accuracy, and improving structural integrity of the bare battery cell.

(10) According to one or more embodiments, the first drive mechanism further includes a first drive motor and a limiting assembly. The limiting assembly includes a connection plate and two limiting plates. The connection plate is connected to the first drive motor, and the two limiting plates are connected to the connection plate. Surfaces of sides of the two limiting plates facing away from the first drive motor are recessed to form limiting grooves. The limiting grooves are configured to accommodate a part of the support plate to limit movement of the support plate relative to the connection plate.

(11) In the foregoing structure, the connection plate is disposed to connect the two limiting plates to the first drive motor, and the limiting plates are provided with the limiting grooves to limit a position of the support plate, thereby reducing a risk that the support plate is displaced during movement, and improving stability in a detection process and accuracy of a detection result.

(12) According to one or more embodiments, a pair of first detection apparatuses is disposed on opposite sides of the cell combining apparatus in alignment with each other, so as to respectively

perform image acquisition on the anode tab back parts of the two bare battery cells disposed in parallel. A pair of second detection apparatuses is disposed on the opposite sides of the cell combining apparatus in alignment with each other, so as to respectively perform image acquisition on the cathode tab back parts of the two bare battery cells disposed in parallel.

(13) Therefore, the detection system of the present disclosure can implement online detection on all the tab back parts of the two bare battery cells disposed in parallel, so as to respectively obtain high-quality images, thereby obtaining more comprehensive information of the quality of the tab back.

(14) According to one or more embodiments, the first detection apparatus and the second detection apparatus each further include a light source. The light source includes a coaxial light source disposed coaxially with the image acquisition mechanism and a first strip-shaped light source and a second strip-shaped light source disposed on two sides of the coaxial light source. The control apparatus is capable of controlling to turn on the coaxial light source, to enable the image acquisition mechanism to acquire an image having a first exposure. Alternatively, the control apparatus is capable of controlling to turn on the coaxial light source, the first strip-shaped light source, and the second strip-shaped light source at the same time, to enable the image acquisition mechanism to acquire an image having a second exposure.

(15) Therefore, the arrangement and illumination combination of the coaxial light source and the strip-shaped light sources may provide optional sufficient supplementary light for the image acquisition mechanism, so that the image acquisition mechanism may acquire high-quality images having a desired exposure according to an actual requirement.

(16) According to one or more embodiments, the first detection apparatus further includes a mounting member disposed on the platform. The image acquisition mechanism includes an imaging apparatus and an imaging lens. The imaging apparatus and the imaging lens are disposed on the mounting member. The imaging lens is disposed between the imaging apparatus and the coaxial light source.

(17) In the foregoing structure, the imaging apparatus is disposed to obtain images of the cathode tab back part and the anode tab back part, so as to implement image acquisition. The imaging lens is disposed to collect light from a photographed object, thereby improving an imaging effect of the imaging apparatus, and improving detection accuracy.

(18) According to one or more embodiments, the cell combining apparatus includes a cell combining flipping mechanism and a second drive mechanism. The cell combining flipping mechanism is configured to be driven by the second drive mechanism to gradually flip and displace the body of the bare battery cell from the lying state to the upright cell combining state.

(19) Therefore, the cell combining apparatus easily implements smooth and stable flipping of the bare battery cell.

(20) According to one or more embodiments, the second drive mechanism includes a second drive motor, a fixed base, a rotation block, and a rotation bearing. The second drive motor and the fixed base are disposed on the platform, the rotation block is disposed on a side of the fixed base facing away from the second drive motor, and an output shaft of the second drive motor passes through the fixed base to be connected to the rotation block. The second drive motor drives the rotation block to rotate about the output shaft of the second drive motor, and the holding apparatus is connected to the rotation block. The rotation bearing is disposed between the rotation block and the fixed base, so as to connect the rotation block to the fixed base rotatably.

(21) In the foregoing structure, the second drive motor is disposed to drive the rotation block to rotate, and the fixed base can support the output shaft of the second drive motor and the rotation block. The rotation block is configured to drive the holding apparatus to rotate, and the rotation bearing connects the rotation block to the fixed base rotatably, thereby reducing resistance and friction in a rotation process of the rotation block, improving stability in the rotation process to improve accuracy of detection, and reducing structural damage to the bare battery cell in the

rotation process to maintain the structural integrity of the bare battery cell.

(22) According to one or more embodiments, a quantity of the fixed bases is two, and two rotation blocks and two rotation bearings are correspondingly provided. The cell combining flipping mechanism further includes a transmission plate. The transmission plate is connected between the two rotation blocks. The holding apparatus is disposed on the transmission plate.

(23) In the foregoing structure, the two fixed bases and the two rotation blocks are disposed to connect two ends of the transmission plate, thereby improving balance of force on the transmission plate and improving stability in the rotation process.

(24) According to one or more embodiments, the cell combining apparatus further includes a third drive mechanism for driving the holding apparatus. The third drive mechanism is configured to drive the holding apparatus to gradually clamp the body of the bare battery cell in a flipping operation process, until the body of the bare battery cell is completely carried by the holding apparatus when the body of the bare battery cell is in the upright cell combining state.

(25) Therefore, slow flipping of the battery cell and gradual increase of the clamping force reduce as much as possible damage to an anode (cathode) tab located between the body of the bare battery cell and the top cap, for example, crack caused by excessively fast bending of the anode (cathode) tab.

(26) According to one or more embodiments, the holding apparatus includes a clamping portion disposed on the transmission plate. The clamping portion includes two clamping members disposed opposite to each other. A space between the two clamping members is configured for clamping the bare battery cell. According to the foregoing structure, the clamping member is disposed on the transmission plate. The clamping member can move with movement of the transmission plate, and clamp the bare battery cell for flipping. The two clamping members are disposed to clamp the bare battery cell from two sides, thereby improving a clamping force and stability in the flipping process.

(27) According to one or more embodiments, the clamping member includes a clamping rod and a battery cell clamping block. One end of the clamping rod is connected to the transmission plate. The battery cell clamping block is disposed at the other end of the clamping rod. The battery cell clamping block is configured to clamp the bare battery cell and drive the bare battery cell to flip. A surface of the battery cell clamping block facing the bare battery cell is recessed to form a clamping groove. A shape of the clamping groove matches a shape of the surface of the bare battery cell.

(28) In the foregoing structure, the clamping rod is disposed to apply a clamping force to the bare battery cell with movement of the transmission plate, and to transfer the clamping force to the bare battery cell by using the battery cell clamping block. The clamping groove is provided on the battery cell clamping block to reduce deformation caused by the clamping force on the surface of the bare battery cell, thereby protecting the surface of the bare battery cell, and improving structural integrity of a detected battery.

(29) According to one or more embodiments, the third drive mechanism includes a third drive motor. The third drive motor is disposed on the transmission plate. The clamping rod is connected to the third drive motor. In the foregoing structure, the third drive motor is disposed to transfer a drive force to the clamping rod, thereby driving the clamping rod to move, and driving the clamping member to clamp the bare battery cell. The clamping force of the clamping member can be accurately adjusted by controlling the third drive motor, and the clamping force for the bare battery cell is increased as a flipping angle increases, to improve stability in the flipping process of the bare battery cell.

(30) According to one or more embodiments, the holding apparatus further includes a drive connection plate. The drive connection plate is connected to the third drive motor. The clamping rod is disposed on the drive connection plate. There are two clamping portions, namely, an upper clamping portion and a lower clamping portion. The upper clamping portion and the lower

clamping portions are spaced apart on the drive connection plate. In the foregoing structure, a plurality of clamping portions are disposed to add clamping points of the bare battery cell, thereby improving clamping stability, and also improving clamping stability for the battery cell in the flipping process.

(31) According to one or more embodiments, a surface of a side of the transmission plate facing the drive connection plate is recessed to form a limiting slot. The limiting slot extends along an arrangement direction of the two clamping members. A surface of a side of the drive connection plate facing the limiting slot protrudes to form a limiting portion. The limiting portion is disposed in the limiting slot. The limiting slot is configured to limit movement of the drive connection plate relative to the transmission plate. In the foregoing structure, the limiting slot extends along the arrangement direction of the two clamping members, so that a movement direction of the drive connection plate can be limited to a direction towards the other clamping member or a direction away from the other clamping member, thereby reducing damage caused to an outer surface of the bare battery cell in a process of moving the clamping members. In addition, the limiting slot is provided, and the limiting portion is disposed in the limiting slot, so that a movement path of the limiting portion can be limited, thereby preventing damage to the bare battery cell caused by an excessively large movement range of the clamping member. The foregoing structure improves stability in the detection process, and enhances protection on the detected bare battery cell.

(32) According to one or more embodiments, the holding apparatus further includes a cushion block. The cushion block is disposed on a side of the transmission plate facing the bare battery cell. In the foregoing structure, the cushion block is disposed to support a side surface of the bare battery cell, thereby improving stability in the flipping process.

(33) In another aspect, the present disclosure provides a detection method. The detection method detects a tab back of a bare battery cell in a cell combining operation process of the bare battery cell by using a detection system. The detection system includes a top cap carrying apparatus, a cell combining apparatus, a first detection apparatus, a second detection apparatus, and a control apparatus. The top cap carrying apparatus is configured to be movable between a carrying position that supports a top cap of a bare battery cell and a displacement position that exactly abuts against but no longer supports the top cap. The method includes: a cell combining operation step: flipping, by using the cell combining apparatus, bodies of two bare battery cells disposed in parallel from a lying state to an upright cell combining state in a case that the top cap of the bare battery cell is supported by the top cap carrying apparatus at the carrying position, where in the upright cell combining state, the body of the bare battery cell is completely held by a holding apparatus and a tab back faces outward; a top cap carrying apparatus moving step: driving, by a first drive mechanism under the control of the control apparatus, the top cap carrying apparatus to move from the carrying position to the displacement position, to allow an anode tab and a cathode tab to extend to expose an anode tab back part and a cathode tab back part within a desired range; an image acquisition step: respectively performing image acquisition on the anode tab back part and the cathode tab back part laterally by using the first detection apparatus and the second detection apparatus; an image processing step: receiving, by using the control apparatus, images acquired by the first detection apparatus and the second detection apparatus, and performing image processing; and a determining step: determining, according to the processed images, whether defects exist in the anode tab back part and the cathode tab back part, for use in a subsequent process.

(34) Therefore, the detection method of the present disclosure can implement online detection on the anode tab back part and the cathode tab back part, so as to obtain more comprehensive information of quality of the tab back, for use in a subsequent process.

(35) According to one or more embodiments, a pair of first detection apparatuses is disposed on opposite sides of the cell combining apparatus in alignment with each other. A pair of second detection apparatuses is disposed on the opposite sides of the cell combining apparatus in alignment with each other. The method includes the following steps: respectively performing, by

using the pair of first detection apparatuses, image acquisition on the anode tab back parts of the two bare battery cells disposed in parallel; and respectively performing, by using the pair of second detection apparatuses, image acquisition on the cathode tab back parts of the two bare battery cells disposed in parallel.

(36) Therefore, in the detection method of the present disclosure, a desired focal length and a field of view of each detection apparatus are used for implementing online detection on the corresponding tab back parts of the two bare battery cells disposed in parallel, so as to obtain high-quality images reflecting the quality of the tab back.

(37) According to one or more embodiments, the first detection apparatus and the second detection apparatus each include an image acquisition mechanism and a light source. The light source includes a coaxial light source disposed coaxially with the image acquisition mechanism and a first strip-shaped light source and a second strip-shaped light source disposed on two sides of the coaxial light source. The method includes the following steps: controlling, by using the control apparatus, to turn on the coaxial light source, to enable the image acquisition mechanism to acquire an image having a first exposure; or controlling, by using the control apparatus, to turn on the coaxial light source, the first strip-shaped light source, and the second strip-shaped light source at the same time, to enable the image acquisition mechanism to acquire an image having a second exposure.

(38) Therefore, optional sufficient supplementary light may be provided for the image acquisition mechanism according to an actual requirement, so as to acquire high-quality images having a desired exposure.

(39) According to one or more embodiments, the cell combining apparatus includes a cell combining flipping mechanism and a second drive mechanism. The cell combining operation step includes: driving, by using the second drive mechanism, the cell combining flipping mechanism, and driving the holding apparatus to gradually flip, so that the body of the bare battery cell is displaced from the lying state to the upright cell combining state.

(40) Therefore, according to the cell combining operation step, the body of the bare battery cell is smoothly displaced into the upright cell combining state.

(41) According to one or more embodiments, the cell combining apparatus further includes a third drive mechanism for driving the holding apparatus. The cell combining operation step includes: driving, by using the third drive mechanism, the holding apparatus to gradually clamp the body of the bare battery cell in a flipping operation process, until the body of the bare battery cell is completely carried by the holding apparatus when the body of the bare battery cell is in the upright cell combining state.

(42) Therefore, according to the cell combining operation step, the holding apparatus gradually clamps the body of the bare battery cell, so that a possibility of cracking a tab located between the body of the bare battery cell and the top cap can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and various other advantages and benefits of the present disclosure will become apparent to those of ordinary skill in the art upon reading the following detailed descriptions of the preferred implementations. The accompanying drawings are merely used for illustrating the preferred implementations and are not intended to constitute a limitation on the present disclosure. In addition, in all the accompanying drawings, same parts are indicated by the same reference numerals.

(2) FIG. 1 schematically shows a perspective view of a detection system according to one or more embodiments of the present disclosure;

- (3) FIG. 2 schematically shows a side view of a detection system according to one or more embodiments of the present disclosure;
- (4) FIG. 3 schematically shows a perspective view of the detection system shown in FIG. 1 after a part of the structure is removed to present a top cap carrying apparatus and a first drive mechanism;
- (5) FIG. 4 schematically shows a side view of the detection system shown in FIG. 1 after a part of the structure is removed to present a top cap carrying apparatus and a first drive mechanism;
- (6) FIG. 5 schematically shows a perspective view of the detection system shown in FIG. 1 after a part of the structure is removed to present a detection apparatus and a bare battery cell;
- (7) FIG. 6 schematically shows a side view of the detection system shown in FIG. 1 after a part of the structure is removed to present a detection apparatus and a bare battery cell;
- (8) FIG. 7 schematically shows a perspective view of the detection system shown in FIG. 1 after a part of the structure is removed to present a cell combining apparatus in which a bare battery cell has been flipped to an upright cell combining state; and
- (9) FIG. 8 schematically shows a flowchart of a detection method according to one or more embodiments of the present disclosure.
- (10) Descriptions of reference numerals: **10**—detection system, **20**—platform, **100**—bare battery cell, **110**—body, **112**—anode tab, **114**—cathode tab, **116**—top cap, **200**—cell combining apparatus, **202**—cell combining flipping mechanism, **204**—second drive mechanism, **206**—second drive motor, **208**—fixed base, **210**—rotation block, **212**—rotation bearing, **214**—transmission plate, **216**—limiting slot, **300A**—first detection apparatus, **300B**—second detection apparatus, **310**—image acquisition mechanism, **312**—imaging apparatus, **314**—imaging lens, **320**—coaxial light source, **330**—first strip-shaped light source, **332**—second strip-shaped light source, **340**—mounting member, **400**—top cap carrying apparatus, **402**—support plate, **404**—first drive mechanism, **406**—limiting fixture, **408**—accommodating groove, **410**—first drive motor, **412**—connection plate, **414**—limiting plate, **500**—holding apparatus, **502**—clamping portion, **504**—clamping member, **506**—clamping rod, **508**—battery cell clamping block, **510**—upper clamping portion, **512**—lower clamping portion, **514**—third drive mechanism, **516**—clamping groove, **518**—third drive motor, **520**—drive connection plate, and **522**—cushion block.

DETAILED DESCRIPTION

- (11) Various embodiments of the technical solutions of the present disclosure will be described in detail below with reference to the accompanying drawings. The following embodiments are only used for illustrating the technical solutions of the present disclosure more explicitly, and are thus only interpreted as examples, rather than used to limit the protection scope of the present disclosure.
- (12) Unless otherwise defined, meanings of all technical and scientific terms used in this specification are the same as those usually understood by a person skilled in the art to which the present disclosure belongs. Terms used in this specification are merely intended to describe objectives of specific embodiments, but are not intended to limit the present disclosure. The terms “include/comprise” and “have” and any variations thereof in the specification and claims of the present disclosure are intended to cover non-exclusive inclusions.
- (13) In the descriptions according to the embodiments of the present disclosure, the technical terms “first”, “second”, and the like are only used for distinguishing different objects, and should not be understood as indicating or implying relative importance or implying the number, specific order or primary and secondary relationship of indicated technical features. In the descriptions according to the embodiments of the present disclosure, “a plurality of” means two or more, unless otherwise expressly and specifically defined.
- (14) Reference herein to “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at least one embodiment of the present disclosure. The presence of the phrase in various places in the specification does not necessarily refer to the same embodiment, nor is it an independent or

alternative embodiment that is mutually exclusive from other embodiments. A person skilled in the art explicitly or implicitly understands that the embodiments described in the specification may be combined with other embodiments.

(15) In the descriptions of the embodiments of the present disclosure, the term “and/or” is merely an association to describe associated objects, and means that there are three relationships. For example, A and/or B may mean that A exists alone, A and B exist at the same time, and B exists alone.

(16) In the descriptions of the embodiments of the present disclosure, the directions or positional relationships indicated by the technical terms shown in the drawings are only for the convenience of describing the embodiments of the present disclosure and simplifying the descriptions, rather than indicating or implying that the involved device or element should have a specific orientation or should be configured or operated in the specific orientation, therefore, they cannot be understood as limiting the embodiments of the present disclosure.

(17) In the descriptions of the embodiments of the present disclosure, unless otherwise explicitly specified and limited, the technical terms “mount”, “connection”, and “fix” should be understood in a broad sense. For example, the connection may be a fixed connection, a detachable connection, or an integral connection. Alternatively, the connection may be a direct connection or an indirect connection through an intermediary. A person of ordinary skill in the art may understand specific meanings of the terms in the embodiments of the present disclosure based on specific situations.

(18) A bare battery cell described in this specification is a single electromagnetic battery cell on which a positive electrode and a negative electrode are already disposed on a top surface. The bare battery cell does not include a protective circuit board and a shell, and cannot be directly used. A surface of the bare battery cell opposite to the top surface is referred to as a bottom surface, a pair of side surfaces having larger cross sections is referred to as large side surfaces, and a pair of side surfaces having smaller cross sections is referred to as small side surfaces.

(19) In the foregoing cell combining operation of two bare battery cells, a body of the bare battery cell needs to be “lifted up” by 90 degrees relative to a welded top cap. Consequently, a tab between the body of the bare battery cell and the top cap is prone to breakage due to a bending force in the process of “lifting up” by 90 degrees. Especially, a back part of the tab facing outward is prone to cracking. In the related art, a detection system and method for performing online detection on a combined bare battery cell to detect whether a defect such as cracking exists in a tab back of the bare battery cell.

(20) Based on the foregoing consideration, a cracking status of the tab back needs to be detected online during the cell combining operation, so as to provide quality reference for a subsequent process. In addition, during the cell combining operation, the tab of the bare battery cell is always in a bent state because the tab and the top cap are still supported by the top cap carrying apparatus when the body is in an upright cell combining state. In this way, the cracking status of the entire tab back cannot be detected. In view of this, it is desired that during detection, the tab can at least partially extend to a desired degree, so as to perform comprehensive and complete online detection on the tab. After detection, a subsequent cell combining operation may be continued. Therefore, the present disclosure provides a detection system and a detection method for detecting a tab back of a bare battery cell in a cell combining apparatus.

(21) The detection system and the detection method according to the present disclosure are described below with reference to the accompanying drawings.

(22) FIG. 1 and FIG. 2 respectively schematically show a perspective view and a side view of a detection system according to one or more embodiments of the present disclosure. FIG. 3 and FIG. 4 respectively schematically show a perspective view and a side view of the detection system shown in FIG. 1 after a part of the structure is removed to present a top cap carrying apparatus and a first drive mechanism.

(23) According to one or more embodiments, referring to FIG. 1 and FIG. 2, a detection system **10**

for detecting a tab back of a bare battery cell in a cell combining apparatus **200** is provided. The detection system **10** includes a top cap carrying apparatus **400**, a cell combining apparatus **200**, a first detection apparatus **300A**, a second detection apparatus **300B**, and a control apparatus. The top cap carrying apparatus **400** is movable at least between a carrying position that supports a top cap **116** of a bare battery cell and a displacement position that exactly abuts against but no longer supports the top cap. The top cap **116** of the bare battery cell is welded together with an anode tab **112** and a cathode tab **114**. The cell combining apparatus **200** is configured to displace bodies **110** of two bare battery cells **100** disposed in parallel from a lying state to an upright cell combining state in a case that the top cap **116** of the bare battery cell is supported by the top cap carrying apparatus **400**. In the upright cell combining state, the body **110** of the bare battery cell is completely held by a holding apparatus **500** and a tab back faces outward. The top cap carrying apparatus **400** is configured to move, under a drive of a first drive mechanism **404**, from the carrying position to the displacement position after the bare battery cell **100** is in the upright cell combining state, to allow the anode tab **112** and the cathode tab **114** to extend to expose an anode tab back part and a cathode tab back part within a desired range. The first detection apparatus **300A** and the second detection apparatus **300B** each include an image acquisition mechanism **310**. The first detection apparatus **300A** is configured to perform image acquisition on the anode tab back part laterally. The second detection apparatus **300B** is configured to perform image acquisition on the cathode tab back part laterally. The control apparatus is configured to control the cell combining apparatus **200** to flip the body **110** of the bare battery cell from the lying state to the upright cell combining state, control movement of the top cap carrying apparatus **400**, control image acquisition by the first detection apparatus **300A** and the second detection apparatus **300B**, and receive and process acquired images, to determine whether defects exist in the anode tab back part and the cathode tab back part, for use in a subsequent process.

(24) Referring to FIG. 1 and FIG. 2, in this specification, the term “lying state” means that a large side surface of the bare battery cell **100** is horizontally carried by a body of the holding apparatus **500**. The term “upright cell combining state” means that the large side surface of the bare battery cell is vertically carried relative to a horizontal plane and a top surface of the bare battery cell having the anode tab **112** and the cathode tab **114** faces downward. In this case, a weight of the bare battery cell in the upright cell combining state is completely held by the holding apparatus **500**. The term “tab back part” is an outer tab side surface part for performing online detection after the anode (cathode) tabs of two bare battery cells **100** at least partially extend when the two bare battery cells are placed in the upright cell combining state facing each other. In this case, sides of the two bare battery cells facing each other are referred to as “inner sides”, and sides facing away from each other are referred to as “outer sides”. In this specification, the term “longitudinal direction” is a direction along which a flipping axis of the cell combining apparatus **200** disposed on a platform **20** of the detection system **10** extends, and a direction parallel to the platform **20** and perpendicular to the flipping axis is referred to as a “lateral direction”. A direction that is perpendicular to the platform **20**, faces upward, and is perpendicular to the flipping axis is referred to as an “upward direction”, and a direction opposite to the “upward direction” is referred to as a “downward direction”.

(25) Referring to FIG. 1 and FIG. 2, the top cap carrying apparatus **400** protrudes from a position below the platform **20** to a position above the platform along the upward direction, and is configured to support the top cap **116** of the bare battery cell **100** in a cell combining operation. It should be noted that in the embodiments of the present disclosure, a structure and layout of the top cap carrying apparatus **400** are not specifically limited, provided that the function thereof can be implemented. The cell combining apparatus **200**, the first detection apparatus **300A**, and the second detection apparatus **300B** are all disposed above the platform **20**, and opposite sides of the cell combining apparatus **200** are correspondingly disposed at a distance from the bare battery cell **100**, so as to perform image acquisition on the anode (cathode) tab back part of the bare battery cell **100**.

It should be noted that in the embodiments of the present disclosure, structures and layouts of the cell combining apparatus **200**, the first detection apparatus **300A**, and the second detection apparatus **300B** are not specifically limited, provided that the function thereof can be implemented. Before the flipping operation in which the cell combining apparatus **200** displaces the body of the bare battery cell **100** from the lying state to the upright cell combining state, the large side surfaces of the bodies **110** of the two bare battery cells disposed in parallel are horizontally carried by the body of the holding apparatus **500** to be in the lying state. The anode tab **112** and the cathode tab **114** of the bare battery cell are both welded with the top cap **116** of the bare battery cell together, and the top cap **116** is supported by the top cap carrying apparatus **400**. In other words, the top cap **116** and the body **110** of the bare battery cell **100** are initially two separate components, and both the anode tab **112** and the cathode tab **114** of the bare battery cell **100** are welded between the top cap **116** and the body **110** in a straight extending manner. When the flipping operation is performed on the bare battery cell, the top cap **116** is always supported by the top cap carrying apparatus **400**, and the body **110** is moved from the lying state to the upright cell combining state. In addition, during the flipping operation, as the cell combining apparatus **200** flips the bare battery cell, the body **110** of the bare battery cell held by the holding apparatus **500** slowly performs circular motion, so that the bodies **110** of the two bare battery cells gradually approach each other, and a clamping force of the holding apparatus **500** is gradually increased, until the clamping force of the holding apparatus is large enough to bear the entire weight of the bare battery cell when the body **110** of the bare battery cell is in the upright cell combining state. In other words, in this case, the body of the holding apparatus **500** is completely adhered to the body of the bare battery cell, and there is a certain moment, so as to prevent the body of the bare battery cell from tipping over. During the circular motion of the body of the bare battery cell, because the top cap **116** is always supported by the top cap carrying apparatus **400**, the anode tab **112** and the cathode tab **114** located between the body **110** of the bare battery cell and the top cap **116** are bent due to 90-degree flipping of the body of the bare battery cell, which may cause forced cracking of a surface layer of the tab back facing outward, thereby causing a defective bare battery cell. In this case, the bare battery cell needs to be detected online.

(26) Referring to FIG. 3 and FIG. 4, the top cap carrying apparatus **400** is movable. Specifically, the top cap carrying apparatus **400** is liftable. The top cap carrying apparatus **400** is configured to be movable at least between the carrying position and the displacement position. In this way, the top cap carrying apparatus **400** may provide an optional and desired detection range for an anode (cathode) tab. At the carrying position, a support plate **402** of the top cap carrying apparatus **400** supports the top cap **116** of the bare battery cell **100**. At the displacement position, the support plate **402** of the top cap carrying apparatus **400** exactly abuts against but no longer supports the top cap **116**, thereby allowing the anode tab **112** and the cathode tab **114** of the bare battery cell **100** to fully extend, for example, extend to be in a natural extension state. Certainly, the top cap carrying apparatus **400** may be moved beyond the carrying position and the displacement position, provided that online detection is not affected. The top cap carrying apparatus **400** may be driven by the first drive mechanism **404** operatively connected thereto to move (for example, move along the downward direction as shown in the figure) from the carrying position to the displacement position away from the body **110** of the bare battery cell by a distance, to allow the anode tab **112** and the cathode tab **114** of the bare battery cell **100** to at least partially extend, thereby providing a desired detection range. Because the weight of the bare battery cell **100** in the upright cell combining state is entirely held by the holding apparatus **500**, in this case, the top cap carrying apparatus **400** abutting against the top cap **116** is moved downward by a distance to at least partially release an upward support force for the bent anode tab **112** and cathode tab **114** and the top cap **116**, and allow the anode tab **112** and the cathode tab **114** to extend downward to expose the anode tab back part and the cathode tab back part within a desired range. The desired range of the anode (cathode) tab back part depends on desired downward extension of the anode (cathode) tab and/or downward

movement of the top cap carrying apparatus.

(27) The first detection apparatus **300A** and the second detection apparatus **300B** each include an image acquisition mechanism **310**, and may be configured to perform image acquisition on the anode tab back part and the cathode tab back part laterally. In the shown embodiment, the image acquisition mechanism **310** is disposed on the platform **20** by using a mounting member. The image acquisition mechanism **310** is suitable for performing high-quality image acquisition on the anode (cathode) tab back part of the bare battery cell **100** laterally. In other words, particularly referring to FIG. **6**, a vertical distance of the image acquisition mechanism **310** relative to the bare battery cell **100** is set to be suitable for enabling the anode (cathode) tab back part of the bare battery cell to be located at a center of a field of view of the image acquisition mechanism **310**. A horizontal distance between the image acquisition mechanism **310** and the bare battery cell **100** is suitable for enabling the bare battery cell **100** to be within a focal length range of the image acquisition mechanism **310**, so as to obtain a clear image.

(28) The detection system **10** of the present disclosure is further provided with the control apparatus (not shown), to control operations of the cell combining apparatus **200**, the first detection apparatus **300A**, the second detection apparatus **300B**, the top cap carrying apparatus **400**, and the like. Specifically, the control apparatus may be configured to control the cell combining apparatus **200** to flip the body of the bare battery cell **100** from the lying state to the upright cell combining state. The control apparatus may be configured to control the first detection apparatus **300A** and the second detection apparatus **300B** to acquire images of the anode tab back part and the cathode tab back part, receive the images acquired by the first detection apparatus **300A** and the second detection apparatus **300B**, and perform image processing, to determine whether a defect exists in the tab back, for use in a subsequent process. It should be noted that in the embodiments of the present application, a structure and layout of the control apparatus are not specifically limited, provided that the function thereof can be implemented. For a subsequent process, for example, a bare battery cell determined to be qualified may be delivered to a next process, or may be temporarily stored into a warehouse, for use in subsequent production. In addition, a bare battery cell determined to be unqualified may be directly removed out of a production line as a waste product, or may be temporarily marked, and is left to be discarded in a subsequent process.

(29) Therefore, the detection system of the present disclosure implements online detection on the anode tab back part and the cathode tab back part of the bare battery cell by using the mobility of the top cap carrying apparatus, thereby obtaining more comprehensive information of quality of the tab back.

(30) According to one or more embodiments, referring to FIG. **1** and FIG. **2**, a pair of first detection apparatuses **300A** is disposed on opposite sides of the cell combining apparatus **200** in alignment with each other, so as to respectively perform image acquisition on the anode tab back parts of the two bare battery cells disposed in parallel. A pair of second detection apparatuses **300B** is disposed on the opposite sides of the cell combining apparatus in alignment with each other, so as to respectively perform image acquisition on the cathode tab back parts of the two bare battery cells disposed in parallel. Therefore, a high-quality image may be acquired from each tab back part. In other words, a total of four tabs of the two bare battery cells disposed in parallel each are provided with a detection apparatus. According to a requirement, the control apparatus may simultaneously or separately control image acquisition operations of the pair of first detection apparatuses **300A** and the pair of second detection apparatuses **300B**.

(31) Therefore, the detection system of the present disclosure can implement online detection on all the tab back parts of the two bare battery cells disposed in parallel, so as to respectively obtain high-quality images, thereby obtaining more comprehensive information of the quality of the tab back.

(32) According to one or more embodiments, the top cap carrying apparatus **400** includes the support plate **402** and the first drive mechanism **404**. The support plate **402** is configured to support

the top cap of the bare battery cell, and the first drive mechanism **404** is disposed on the platform **20**. The first drive mechanism **404** is connected to the support plate **402**. The first drive mechanism **404** is connected to the support plate **402**, and is capable of driving the support plate **402** to move in a direction from the bare battery cell **100** to the platform **20**, so that the anode tab **112** and the cathode tab **114** extend.

(33) In the foregoing structure, the first drive mechanism **404** can provide a drive force for movement of the support plate **402**, and the support plate **402** can transfer the drive force of the first drive mechanism **404** to the top cap of the bare battery cell **100**. By adjusting a movement distance of the support plate **402**, the top cap of the bare battery cell **100** may be moved between the top cap carrying position and the displacement position, to adjust extension states of the anode tab and the cathode tab.

(34) In one embodiment, a distance between the top cap of the bare battery cell **100** and the first drive mechanism **404** can be controlled by adjusting a length of the support plate **402**, thereby effectively improving adaptability of the first drive mechanism **404** to different bare battery cells.

(35) According to one or more embodiments, the first drive mechanism **404** further includes a limiting fixture **406**. The limiting fixture **406** is disposed at an end of the support plate **402** facing the bare battery cell **100**. A surface of a side of the limiting fixture **406** facing the bare battery cell is recessed to form an accommodating groove **408**. A shape of the accommodating groove **408** matches that of the top cap of the bare battery cell.

(36) In the foregoing structure, the limiting fixture **406** can guide a movement direction of the top cap of the bare battery cell, thereby reducing a risk that the top cap of the bare battery cell is deformed and displaced in a flipping process, improving detection accuracy, and improving structural integrity of the bare battery cell.

(37) According to one or more embodiments, the first drive mechanism **404** further includes a first drive motor **410** and a limiting assembly. The limiting assembly includes a connection plate **412** and two limiting plates **414**. The connection plate **412** is connected to the first drive motor **410**, and the two limiting plates **414** are connected to the connection plate **412**. Surfaces of sides of the two limiting plates **414** facing away from the first drive motor **410** are recessed to form limiting grooves. The limiting grooves are configured to accommodate a part of the support plate **401** to limit movement of the support plate **401** relative to the connection plate **412**.

(38) In the foregoing structure, the connection plate **412** is disposed to connect the two limiting plates **414** to the first drive motor **410**, and the limiting plates **414** are provided with the limiting grooves to limit a position of the support plate **402**, thereby reducing a risk that the support plate **402** is displaced during movement, and improving stability in a detection process and accuracy of a detection result.

(39) FIG. 5 and FIG. 6 respectively schematically show a perspective view and a side view of the detection system shown in FIG. 1 after a part of the structure is removed to present a first detection apparatus, a second detection apparatus, and a bare battery cell.

(40) According to one or more embodiments, referring to FIG. 5 and FIG. 6, the first detection apparatus **300A** and the second detection apparatus **300B** each further include a light source. The light source includes a coaxial light source **320** disposed coaxially with the image acquisition mechanism **310** and a first strip-shaped light source **330** and a second strip-shaped light source **332** disposed on two sides of the coaxial light source. The control apparatus is capable of controlling to turn on the coaxial light source **320**, to enable the image acquisition mechanism **310** to acquire an image having a first exposure. Alternatively, the control apparatus is capable of controlling to turn on the coaxial light source **320**, the first strip-shaped light source **330**, and the second strip-shaped light source **332** at the same time, to enable the image acquisition mechanism **310** to acquire an image having a second exposure.

(41) Referring to FIG. 1, FIG. 5, and FIG. 6, the first detection apparatus **300A** and the second detection apparatus **300B** each include the image acquisition mechanism **310**, the coaxial light

source **320**, the first strip-shaped light source **330**, and the second strip-shaped light source **332**.

(42) The light source may include the coaxial light source **320**, the first strip-shaped light source **330**, and the second strip-shaped light source **332**. Referring to FIG. 5, the coaxial light source **320** is disposed between the image acquisition mechanism **310** and the bare battery cell **100**. It should be noted that in the embodiments of the present application, a specific structure of the coaxial light source **320** is not limited, provided that a central axis of the coaxial light source **320** and an optical axis of the image acquisition mechanism **310** are disposed on a same axis. The coaxial light source **320** may provide straight and uniform supplementary light for the image acquisition mechanism **310**. The two strip-shaped light sources **330** are disposed between the image acquisition mechanism **310** and the bare battery cell **100** and on both sides of the coaxial light source **320**. The two strip-shaped light sources **330** may provide supplementary light converging from two lateral directions to the image acquisition mechanism **310**. It should be noted that in the embodiments of the present application, a specific structure of the strip-shaped light source **330** is not limited, and a light source having any geometric shape may be used.

(43) The control apparatus may control to turn on the light sources to provide supplementary light for the image acquisition mechanism **310**, so as to obtain high-quality images. When the image acquisition mechanism **310** performs image acquisition, the control apparatus may selectively control illumination of the light sources, so that only the coaxial light source **320** illuminates, to enable the image acquisition mechanism to acquire an image having a first exposure. Alternatively, the coaxial light source and the two strip-shaped light sources **330** simultaneously illuminate, to enable the image acquisition mechanism to acquire an image having a second exposure. Therefore, the arrangement and illumination combination of the coaxial light source and the strip-shaped light sources may provide optional sufficient supplementary light for the image acquisition mechanism, so that the image acquisition mechanism may acquire high-quality images having a desired exposure according to an actual requirement.

(44) According to one or more embodiments, the first detection apparatus **300A** further includes a mounting member **340** disposed on the platform **20**. The image acquisition mechanism includes an imaging apparatus **312** and an imaging lens **314**. The imaging apparatus **312** and the imaging lens **314** are disposed on the mounting member **340**. The imaging lens **314** is disposed between the imaging apparatus **312** and the coaxial light source **320**. Exemplarily, the imaging apparatus **312** may be a digital camera or a video camera. The imaging lens **314** may be a lens for focusing or improving photographing precision, which matches the imaging apparatus **312**.

(45) In the foregoing structure, the imaging apparatus **312** is disposed to obtain images of the back parts of the cathode tab **114** and the anode tab **112**, so as to implement image acquisition. The imaging lens **314** is disposed to collect light from a photographed object, thereby improving an imaging effect of the imaging apparatus **312**, and improving detection accuracy.

(46) FIG. 7 schematically shows a perspective view of the detection system shown in FIG. 1 after a part of the structure is removed to present a cell combining apparatus in which a bare battery cell has been flipped to an upright cell combining state.

(47) According to one or more embodiments, referring to FIG. 1 and FIG. 7, the cell combining apparatus **200** includes a cell combining flipping mechanism **202** and a second drive mechanism **204**. The cell combining flipping mechanism **202** is configured to be driven by the second drive mechanism **204** to gradually flip and displace the body of the bare battery cell **100** from the lying state to the upright cell combining state.

(48) It should be noted that in the embodiments of the present application, a specific structure of the cell combining apparatus **200** is not limited, provided that the function thereof can be implemented. Referring to FIG. 1 and FIG. 7, under the control of the control apparatus, the cell combining apparatus **200** can be driven by the second drive mechanism **204** to perform a flipping operation on the bare battery cell **100**, so that the body of the bare battery cell is gradually flipped by 90 degrees from the lying state along a circular path to be displaced to the upright cell

combining state.

(49) Therefore, the cell combining apparatus easily implements smooth and stable flipping of the bare battery cell.

(50) According to one or more embodiments, the second drive mechanism **204** includes a second drive motor **206**, a fixed base **208**, a rotation block **210**, and a rotation bearing **212**. The second drive motor **206** and the fixed base **208** are disposed on the platform **20**. The rotation block **210** is disposed on a side of the fixed base **208** facing away from the second drive motor **206**. An output shaft of the second drive motor **206** passes through the fixed base **208** to be connected to the rotation block **210**. The second drive motor **206** drives the rotation block **201** to rotate about the output shaft of the second drive motor, and the holding apparatus is connected to the rotation block. The rotation bearing **212** is disposed between the rotation block **210** and the fixed base **208**, so as to connect the rotation block **210** to the fixed base **208** rotatably.

(51) In the foregoing structure, the second drive motor **206** is disposed to drive the rotation block **210** to rotate, and the fixed base **208** can support the output shaft of the second drive motor **206** and the rotation block **210**. The rotation block **210** is configured to drive the holding apparatus **500** to rotate, and the rotation bearing **212** connects the rotation block **210** to the fixed base **208** rotatably, thereby reducing resistance and friction in a rotation process of the rotation block **210**, improving stability in the rotation process to improve accuracy of detection, and reducing structural damage to the bare battery cell **100** in the rotation process to maintain the structural integrity of the bare battery cell **100**.

(52) According to one or more embodiments, a quantity of the fixed bases **208** is two, and two rotation blocks **210** and two rotation bearings **212** are correspondingly provided. The cell combining flipping mechanism **202** further includes a transmission plate **214**. The transmission plate **214** is connected between the two rotation blocks **210**. The holding apparatus **500** is disposed on the transmission plate **214**.

(53) In the foregoing structure, the two fixed bases **208** and the two rotation blocks **210** are disposed to connect two ends of the transmission plate **214**, thereby improving balance of force on the transmission plate **214** and improving stability in the rotation process.

(54) According to one or more embodiments, referring to FIG. 1 and FIG. 7, the cell combining apparatus **200** further includes a third drive mechanism **514** for driving the holding apparatus **500**. The third drive mechanism is configured to drive the holding apparatus **500** to gradually clamp the body **110** of the bare battery cell in a flipping operation process, until the body of the bare battery cell is completely carried by the holding apparatus **500** when the body of the bare battery cell is in the upright cell combining state.

(55) It should be noted that in the embodiments of the present application, specific structures of the holding apparatus **500** and the third drive mechanism **514** are not limited, provided that the functions thereof can be implemented. Referring to FIG. 7, the holding apparatus **500** may include a pair of upper clamping portions **510** and a pair of lower clamping portions **512**. In a process in which the body of the bare battery cell is flipped from the lying state to the upright cell combining state, the body of the bare battery cell slowly performs circular motion to make the bodies **110** of the two bare battery cells gradually approach each other, and clamping forces provided by the upper clamping portion **510** and the lower clamping portion **512** of the holding apparatus **500** are gradually increased, until the clamping force is large enough to bear the entire weight of the bare battery cell when the body of the bare battery cell is in the upright cell combining state. In this case, the body of the holding apparatus is completely adhered to the body of the bare battery cell, and the upper clamping portion **510** and the lower clamping portion **512** respectively clamp the body of the bare battery cell **100** at an upper part and a lower part from a small side surface and provide a certain moment to the bare battery cell, so as to prevent the bare battery cell **100** from tipping over.

(56) Therefore, slow flipping of the battery cell and gradual increase of the clamping force reduce

as much as possible damage to an anode (cathode) tab located between the body of the bare battery cell and the top cap, for example, crack caused by excessively fast bending of the anode (cathode) tab.

(57) According to one or more embodiments, the holding apparatus **500** includes a clamping portion **502** disposed on the transmission plate. The clamping portion **502** includes two clamping members **504** disposed opposite to each other. A space between the two clamping members **504** is configured for clamping the bare battery cell. According to the foregoing structure, the clamping member **504** is disposed on the transmission plate **214**. The clamping member **504** can move with movement of the transmission plate **214**, and clamp the bare battery cell **100** for flipping. The two clamping members **504** are disposed to clamp the bare battery cell **100** from two sides, thereby improving a clamping force and stability in the flipping process.

(58) According to one or more embodiments, the clamping member **504** includes a clamping rod **506** and a battery cell clamping block **508**. One end of the clamping rod **506** is connected to the transmission plate **214**. The battery cell clamping block **508** is disposed at the other end of the clamping rod **506**. The battery cell clamping block **508** is configured to clamp the bare battery cell **100** and drive the bare battery cell **100** to flip. A surface of the battery cell clamping block **508** facing the bare battery cell **100** is recessed to form a clamping groove **516**. A shape of the clamping groove **516** matches a shape of the surface of the bare battery cell **100**.

(59) In the foregoing structure, the clamping rod **506** is disposed to apply a clamping force to the bare battery cell **100** with movement of the transmission plate **214**, and to transfer the clamping force to the bare battery cell **100** by using the battery cell clamping block **508**. The clamping groove **516** is provided on the battery cell clamping block **508** to reduce deformation caused by the clamping force on the surface of the bare battery cell **100**, thereby protecting the surface of the bare battery cell **100**, and improving structural integrity of a detected battery.

(60) According to one or more embodiments, the third drive mechanism **514** includes a third drive motor **518**. The third drive motor **518** is disposed on the transmission plate **214**. The clamping rod **506** is connected to the third drive motor **518**. In the foregoing structure, the third drive motor **518** is disposed to transfer a drive force to the holding apparatus **500**, thereby driving the clamping rod **506** to move, and driving the clamping member **504** to clamp the bare battery cell. The clamping force of the clamping member **504** can be accurately adjusted by controlling the third drive motor **518**, and the clamping force for the bare battery cell **100** is increased as a flipping angle increases, to improve stability in the flipping process of the bare battery cell **100**.

(61) According to one or more embodiments, the holding apparatus **500** further includes a drive connection plate **520**. The drive connection plate **520** is connected to the third drive motor **518**. The clamping rod **506** is disposed on the drive connection plate **520**. There are two clamping portions **502**, namely, an upper clamping portion **510** and a lower clamping portion **512**. The upper clamping portion **510** and the lower clamping portion **512** are spaced apart on the drive connection plate **520**.

(62) In the foregoing structure, a plurality of clamping portions **502** are disposed to add clamping points of the bare battery cell **100**, thereby improving clamping stability, and also improving clamping stability for the battery cell in the flipping process.

(63) According to one or more embodiments, a surface of a side of the transmission plate **214** facing the drive connection plate **520** is recessed to form a limiting slot **216**. The limiting slot **216** extends along an arrangement direction of the two clamping members **504**. It may be understood that the two clamping members **504** herein are two clamping members **504** disposed on both sides of the bare battery cell **100** in a same clamping portion **502**, and a space for clamping the bare battery cell **100** is formed between the two clamping members **504**. A surface of a side of the drive connection plate **520** facing the limiting slot **216** protrudes to form a limiting portion. The limiting portion is disposed in the limiting slot **216**. The limiting slot **216** is configured to limit movement of the drive connection plate **520** relative to the transmission plate **214**. In the foregoing structure,

the limiting slot **216** extends along the arrangement direction of the two clamping members **504**, so that a movement direction of the drive connection plate **520** can be limited to a direction towards the other clamping member **504** or a direction away from the other clamping member **504**, thereby reducing damage caused to an outer surface of the bare battery cell **100** in a process of moving the clamping members **504**. In addition, the limiting slot **216** is provided, and the limiting portion is disposed in the limiting slot **216**, so that a movement path of the limiting portion can be limited, thereby preventing damage to the bare battery cell **100** caused by an excessively large movement range of the clamping member **504**. The foregoing structure improves stability in the detection process, and enhances protection on the detected bare battery cell **100**.

(64) According to one or more embodiments, the holding apparatus **500** further includes a cushion block **522**. The cushion block **522** is disposed on a side of the transmission plate **214** facing the bare battery cell **100**. In the foregoing structure, the cushion block **522** is disposed to support a side surface of the bare battery cell **100**, thereby improving stability in the flipping process. In one embodiment, a thickness of the cushion block **522** may be adjusted according to bare battery cells **100** having different thicknesses, to adapt to testing of battery cells having different specifications.

(65) According to one or more embodiments, referring to FIG. **8**, a detection method is provided. The detection method detects a tab back part of a bare battery cell **100** in a cell combining operation process of the bare battery cell by using a detection system **10**. The detection system includes a top cap carrying apparatus **400**, a cell combining apparatus **200**, a first detection apparatus **300A**, a second detection apparatus **300B**, and a control apparatus. The top cap carrying apparatus **400** is configured to move between a carrying position that supports a top cap of the bare battery cell and a displacement position that exactly abuts against but no longer supports the top cap. The method includes a cell combining operation step **S1**, a top cap carrying apparatus moving step **S2**, an image acquisition step **S3**, an image processing step **S4**, and a determining step **S5**. In the cell combining operation step **S1**, bodies of two bare battery cells **100** disposed in parallel are flipped from a lying state to an upright cell combining state by using the cell combining apparatus **200** in a case that the top cap of the bare battery cell is supported by the top cap carrying apparatus **400**, where in the upright cell combining state, the body of the bare battery cell is completely held by a holding apparatus **500** and a tab back faces outward. In the top cap carrying apparatus moving step **S2**, a first drive mechanism **404** drives, under the control of the control apparatus, the top cap carrying apparatus **400** to move from the carrying position to the displacement position, to allow an anode tab and a cathode tab to extend to expose an anode tab back part and a cathode tab back part within a desired range. In the image acquisition step **S3**, the first detection apparatus **300A** and the second detection apparatus **300B** respectively perform image acquisition on the anode tab back part and the cathode tab back part laterally. In the image processing step **S4**, the control apparatus receives images acquired by the first detection apparatus **300A** and the second detection apparatus **300B**, and performs image processing. In the determining step, according to the processed images, whether defects exist in the tab back parts is determined, for use in a subsequent process.

(66) FIG. **8** shows a flowchart of a detection method according to the present disclosure. The detection method according to the present disclosure may be performed by using the detection system **10** according to the present disclosure. The detection method includes the following steps: the cell combining operation step **S1**, the top cap carrying apparatus moving step **S2**, the image acquisition step **S3**, the image processing step **S4**, and the determining step **S5**. After detection is completed, the cell combining apparatus **200** may continue to perform a subsequent cell combining operation.

(67) Therefore, the detection method of the present disclosure can implement online detection on the anode tab back part and the cathode tab back part, so as to obtain more comprehensive information of quality of the tab back, so that the acquired and processed image can be used for a subsequent process.

(68) According to one or more embodiments, a pair of first detection apparatuses **300A** is disposed

on opposite sides of the cell combining apparatus in alignment with each other. A pair of second detection apparatuses **300B** is disposed on the opposite sides of the cell combining apparatus in alignment with each other. The method includes the following steps: respectively performing, by using the pair of first detection apparatuses, image acquisition on the anode tab back parts of the two bare battery cells disposed in parallel; and respectively performing, by using the pair of second detection apparatuses, image acquisition on the cathode tab back parts of the two bare battery cells disposed in parallel.

(69) Specifically, detection apparatuses are respectively provided for four tabs of the two bare battery cells disposed in parallel. In addition, a horizontal distance and a vertical distance of each image acquisition mechanism relative to the bare battery cell **100** are set to obtain a desired focal length and field of view, so as to acquire a high-quality image from the anode tab back part or the cathode tab back part of the corresponding bare battery cell **100**.

(70) In the detection method of the present disclosure, a desired focal length and a field of view of each detection apparatus are used for implementing online detection on the corresponding tab back parts of the two bare battery cells disposed in parallel, so as to obtain high-quality images reflecting the quality of the tab back.

(71) According to one or more embodiments, the first detection apparatus **300A** and the second detection apparatus **300B** each include an image acquisition mechanism **310** and a light source. The light source includes a coaxial light source **320** disposed coaxially with the image acquisition mechanism **310** and a first strip-shaped light source **330** and a second strip-shaped light source **332** disposed on two sides of the coaxial light source. The method includes the following steps: controlling, by using the control apparatus, to turn on the coaxial light source **320**, to enable the image acquisition mechanism **310** to acquire an image having a first exposure; or controlling, by using the control apparatus, to turn on the coaxial light source **320**, the first strip-shaped light source **330**, and the second strip-shaped light source **332** at the same time, to enable the image acquisition mechanism **310** to acquire an image having a second exposure.

(72) Therefore, optional sufficient supplementary light may be provided for the image acquisition mechanism according to an actual requirement, so as to acquire high-quality images having a desired exposure.

(73) According to one or more embodiments, the cell combining apparatus **200** includes a cell combining flipping mechanism **202** and a second drive mechanism **204**. The cell combining operation step **S1** includes: driving, by using the second drive mechanism **204**, the cell combining flipping mechanism **202**, and driving the holding apparatus **500** to gradually flip, so that the body of the bare battery cell **100** is displaced from the lying state to the upright cell combining state.

(74) Specifically, by using the detection system according to the present disclosure, the second drive mechanism **204** drives the cell combining flipping mechanism **202**, and the body of the bare battery cell held by the holding apparatus **500** may slowly perform circular motion from the lying state to the upright cell combining state.

(75) Therefore, according to the cell combining operation step, the body of the bare battery cell is smoothly displaced into the upright cell combining state.

(76) According to one or more embodiments, the cell combining apparatus **200** further includes a third drive mechanism **514** for driving the holding apparatus **500**. The cell combining operation step **S1** includes: driving, by using the third drive mechanism **514**, the holding apparatus **500** to gradually clamp the bare battery cell **100** in a flipping operation process, until the body of the bare battery cell is completely carried by the holding apparatus when the body of the bare battery cell is in the upright cell combining state.

(77) Specifically, by using the detection system according to the present disclosure, the holding apparatus **500** is driven through the third drive mechanism **514**, so that the clamping force of the holding apparatus **500** is gradually increased during the flipping operation, until the clamping force of the holding apparatus is large enough to bear the entire weight of the bare battery cell **100** when

the body of the bare battery cell is in the upright cell combining state.

(78) Therefore, according to the cell combining operation step, the holding apparatus gradually clamps the body of the bare battery cell, so that a possibility of cracking a tab located between the body of the bare battery cell and the top cap can be reduced.

(79) Finally, it should be noted that the foregoing embodiments are merely used for describing the technical solutions of the present disclosure, but are not intended to limit the technical solutions. Although the present disclosure is described in detail with reference to the foregoing embodiments, it should be appreciated by a person skilled in the art that modifications may still be made to the technical solutions described in the foregoing embodiments, or equivalent replacements may be made to part or all of the technical features. However, these modifications or replacements do not cause the essence of the corresponding technical solutions to deviate from the scope of the technical solutions of the embodiments of the present disclosure, and should be covered by the scope of the claims and the specification of the present disclosure. Especially, as long as there is no structural conflict, the various technical features mentioned in the embodiments may be combined in any way. The present disclosure is not limited to the particular embodiments disclosed herein, but includes all technical solutions that fall within the scope of the claims.

Claims

1. A detection system for detecting a tab back of a bare battery cell in a cell combining apparatus, the detection system comprising: a top cap carrying apparatus, the top cap carrying apparatus being configured to be movable at least between a carrying position that supports a top cap of a bare battery cell and a displacement position that exactly abuts against but no longer supports the top cap, the top cap of the bare battery cell being welded together with an anode tab and a cathode tab; a cell combining apparatus, the cell combining apparatus being configured to displace bodies of two bare battery cells disposed in parallel from a lying state to an upright cell combining state in a case that the top cap is supported by the top cap carrying apparatus, in the upright cell combining state, the body of the bare battery cell being completely held by a holding apparatus and a tab back facing outward, wherein the top cap carrying apparatus is configured to move, under a drive of a first drive mechanism, from the carrying position to the displacement position after the bare battery cell is in the upright cell combining state, to allow the anode tab and the cathode tab to extend to expose an anode tab back part and a cathode tab back part within a desired range; a first detection apparatus and a second detection apparatus, both the first detection apparatus and the second detection apparatus comprising an image acquisition mechanism, the first detection apparatus being configured to perform image acquisition on the anode tab back part laterally, and the second detection apparatus being configured to perform image acquisition on the cathode tab back part laterally; and a control apparatus, the control apparatus being configured to control the cell combining apparatus to flip the body of the bare battery cell from the lying state to the upright cell combining state, control movement of the top cap carrying apparatus, control image acquisition by the first detection apparatus and the second detection apparatus, and receive and process acquired images, to determine whether defects exist in the anode tab back part and the cathode tab back part, for use in a subsequent process.

2. The detection system according to claim 1, wherein the top cap carrying apparatus comprises: a support plate, configured to support the top cap of the bare battery cell; and a first drive mechanism disposed on a platform, the first drive mechanism being connected to the support plate, and the first drive mechanism being capable of driving the support plate to move in a direction from the bare battery cell to the platform, so that the anode tab and the cathode tab extend.

3. The detection system according to claim 2, wherein the first drive mechanism further comprises a limiting fixture, the limiting fixture is disposed at an end of the support plate facing the bare battery cell, a surface of a side of the limiting fixture facing the bare battery cell is recessed to form

an accommodating groove, and a shape of the accommodating groove matches that of the top cap of the bare battery cell.

4. The detection system according to claim 3, wherein the first drive mechanism further comprises a first drive motor and a limiting assembly, and the limiting assembly comprises: a connection plate, connected to the first drive motor; and two limiting plates, connected to the connection plate, wherein surfaces of sides of the two limiting plates facing away from the first drive motor are recessed to form limiting grooves, and the limiting grooves are configured to accommodate a part of the support plate to limit movement of the support plate relative to the connection plate.

5. The detection system according to claim 1, wherein a pair of first detection apparatuses is disposed on opposite sides of the cell combining apparatus in alignment with each other, so as to respectively perform image acquisition on the anode tab back parts of the two bare battery cells disposed in parallel; and a pair of second detection apparatuses is disposed on the opposite sides of the cell combining apparatus in alignment with each other, so as to respectively perform image acquisition on the cathode tab back parts of the two bare battery cells disposed in parallel.

6. The detection system according to claim 5, wherein the first detection apparatus and the second detection apparatus further comprise a light source, and the light source comprises a coaxial light source disposed coaxially with the image acquisition mechanism and a first strip-shaped light source and a second strip-shaped light source disposed on two sides of the coaxial light source; and the control apparatus is capable of controlling to turn on the coaxial light source, to enable the image acquisition mechanism to acquire an image having a first exposure; or the control apparatus is capable of controlling to turn on the coaxial light source, the first strip-shaped light source, and the second strip-shaped light source at the same time, to enable the image acquisition mechanism to acquire an image having a second exposure.

7. The detection system according to claim 6, wherein the first detection apparatus further comprises a mounting member disposed on the platform, the image acquisition mechanism comprises an imaging apparatus and an imaging lens, the imaging apparatus and the imaging lens are disposed on the mounting member, and the imaging lens is disposed between the imaging apparatus and the coaxial light source.

8. The detection system according to claim 1, wherein the cell combining apparatus comprises a cell combining flipping mechanism and a second drive mechanism, and the cell combining flipping mechanism is configured to be driven by the second drive mechanism to gradually flip and displace the body of the bare battery cell from the lying state to the upright cell combining state.

9. The detection system according to claim 8, wherein the second drive mechanism comprises: a second drive motor, disposed on the platform; a fixed base, disposed on the platform; a rotation block, disposed on a side of the fixed base facing away from the second drive motor, wherein an output shaft of the second drive motor passes through the fixed base to be connected to the rotation block, the second drive motor drives the rotation block to rotate about the output shaft of the second drive motor, and the holding apparatus is connected to the rotation block; and a rotation bearing, disposed between the rotation block and the fixed base, so as to connect the rotation block to the fixed base rotatably.

10. The detection system according to claim 9, wherein a quantity of the fixed bases is two, two rotation blocks and two rotation bearings are correspondingly provided, the cell combining flipping mechanism further comprises a transmission plate, the transmission plate is connected between the two rotation blocks, and the holding apparatus is disposed on the transmission plate.

11. The detection system according to claim 10, wherein the cell combining apparatus further comprises a third drive mechanism for driving the holding apparatus, the third drive mechanism is configured to drive the holding apparatus to gradually clamp the body of the bare battery cell in a flipping operation process, until the body of the bare battery cell is completely carried by the holding apparatus when the body of the bare battery cell is in the upright cell combining state.

12. The detection system according to claim 11, wherein the holding apparatus comprises a

clamping portion disposed on the transmission plate, the clamping portion comprises two clamping members disposed opposite to each other, and a space between the two clamping members is configured for clamping the bare battery cell.

13. The detection system according to claim 12, wherein the clamping member comprises: a clamping rod, one end of the clamping rod being connected to the transmission plate; and a battery cell clamping block disposed at the other end of the clamping rod, wherein the battery cell clamping block is configured to clamp the bare battery cell and drive the bare battery cell to flip, a surface of the battery cell clamping block facing the bare battery cell is recessed to form a clamping groove, and a shape of the clamping groove matches a shape of the surface of the bare battery cell.

14. The detection system according to claim 13, wherein the third drive mechanism comprises a third drive motor, the third drive motor is disposed on the transmission plate, and the clamping rod is connected to the third drive motor.

15. The detection system according to claim 14, wherein the holding apparatus further comprises a drive connection plate, the drive connection plate is connected to the third drive motor, and the clamping rod is disposed on the drive connection plate; and there are two clamping portions, namely, an upper clamping portion and a lower clamping portion, and the upper clamping portion and the lower clamping portion are spaced apart on the drive connection plate.

16. The detection system according to claim 12, wherein a surface of a side of the transmission plate facing the drive connection plate is recessed to form a limiting slot, and the limiting slot extends along an arrangement direction of the two clamping members; and a surface of a side of the drive connection plate facing the limiting slot protrudes to form a limiting portion, the limiting portion is disposed in the limiting slot, and the limiting slot is configured to limit movement of the drive connection plate relative to the transmission plate.

17. The detection system according to claim 10, wherein the holding apparatus further comprises a cushion block, and the cushion block is disposed on a side of the transmission plate facing the bare battery cell.

18. A detection method, the detection method detecting a tab back of a bare battery cell in a cell combining operation process of the bare battery cell by using a detection system, the detection system comprising a top cap carrying apparatus, a cell combining apparatus, a first detection apparatus, a second detection apparatus, and a control apparatus, and the top cap carrying apparatus being configured to be movable between a carrying position that supports a top cap of the bare battery cell and a displacement position that exactly abuts against but no longer supports the top cap, wherein the detection method comprises: a cell combining operation step: flipping, by using the cell combining apparatus, bodies of two bare battery cells disposed in parallel from a lying state to an upright cell combining state in a case that the top cap of the bare battery cell is supported by the top cap carrying apparatus at the carrying position, in the upright cell combining state, the body of the bare battery cell being completely held by a holding apparatus and a tab back facing outward; a top cap carrying apparatus moving step: driving, by a first drive mechanism under the control of the control apparatus, the top cap carrying apparatus to move from the carrying position to the displacement position, to allow an anode tab and a cathode tab to extend to expose an anode tab back part and a cathode tab back part within a desired range; an image acquisition step: respectively performing image acquisition on the anode tab back part and the cathode tab back part laterally by using the first detection apparatus and the second detection apparatus; an image processing step: receiving, by using the control apparatus, images acquired by the first detection apparatus and the second detection apparatus, and performing image processing; and a determining step: determining, according to the processed images, whether defects exist in the anode tab back part and the cathode tab back part, for use in a subsequent process; the cell combining apparatus comprises a cell combining flipping mechanism and a second drive mechanism, and the cell combining operation step comprises: driving, by using the second drive mechanism, the cell

combining flipping mechanism, and driving the holding apparatus to gradually flip, so that the body of the bare battery cell is displaced from the lying state to the upright cell combining state; and the cell combining apparatus further comprises a third drive mechanism for driving the holding apparatus, and the cell combining operation step comprises: driving, by using the third drive mechanism, the holding apparatus to gradually clamp the body of the bare battery cell in a flipping operation process, until the body of the bare battery cell is completely carried by the holding apparatus when the body of the bare battery cell is in the upright cell combining state.

19. The detection method according to claim 18, wherein a pair of first detection apparatuses is disposed on opposite sides of the cell combining apparatus in alignment with each other, and a pair of second detection apparatuses is disposed on the opposite sides of the cell combining apparatus in alignment with each other; and the method comprises the following steps: respectively performing, by using the pair of first detection apparatuses, image acquisition on the anode tab back parts of the two bare battery cells disposed in parallel; and respectively performing, by using the pair of second detection apparatuses, image acquisition on the cathode tab back parts of the two bare battery cells disposed in parallel.

20. The detection method according to claim 19, wherein both the first detection apparatus and the second detection apparatus comprise an image acquisition mechanism and a light source, and the light source comprises a coaxial light source disposed coaxially with the image acquisition mechanism and a first strip-shaped light source and a second strip-shaped light source disposed on two sides of the coaxial light source; and the method comprises the following steps: controlling, by using the control apparatus, to turn on the coaxial light source, to enable the image acquisition mechanism to acquire an image having a first exposure; or controlling, by using the control apparatus, to turn on the coaxial light source, the first strip-shaped light source, and the second strip-shaped light source at the same time, to enable the image acquisition mechanism to acquire an image having a second exposure.
